A 256-STEP CERTIFICATE FOR TURÁN'S PURE POWER-SUM CONSTANT

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Abstract. We give an exact-input certificate for the upper bound $C_{42} \le 0.6888863712$ in the pure power-sum problem. The construction refines the step profiles of Griego, Röhrig and Plata to 256 middle blocks, without changing their coefficient-cancellation mechanism. It improves the strongest public proposal located in our 22 October 2026 audit, 0.688970, by 0.0000836288. An unmodified public interval checker and a separately written Arb checker verify the strict finite inequalities. The asymptotic implication is proved below; no explicit finite threshold, optimality, human peer review or formal verification is claimed. All material remains local and unsubmitted.

1. Statement and prior constructions

Define

$$R_n = \min_{\substack{z_1,\dots,z_n\in\mathbb{C}\\ \max_j |z_j|=1}} \max_{1\le k\le n} \left|\sum_{j=1}^{n} z_j^k\right|, \qquad C_{42} = \limsup_{n\to\infty} R_n.$$

This is constant 42a in Tao and contributors' table [1]. The condition is maximum modulus one; the other roots may lie inside the unit disk.

Theorem 1. *The pure power-sum constant satisfies*

$$C_{42} \le \frac{215276991}{312500000} = 0.6888863712.$$

The table lists Harcos's 0.69368 and Griego's proposed 0.6906538 [2, 3]. Open PR 184 proposes Röhrig's eight-step bound 0.688983 [4]. A separate public repository by Kerrigan Plata proposes 0.688970 using six nonuniform middle levels [5]. We reproduced both latter certificates before optimizing. Plata's value is the strongest public upper proposal located, and is the comparison used here; both proposals explicitly lack expert peer review. The local audit is bounded, not an exhaustive priority claim.

We retain the generating function, free final block and Newton-identity argument of these works. The changes are a finer middle profile and new rational inputs. No new power-sum reduction theorem is needed.

2. A finite-step criterion

Put $\tau = 1/3$ and $B = 2/3$. Let

$$\tau = t_0 < t_1 < \cdots < t_M = B, \qquad \alpha = a + ib, \quad a > 0, \quad s = 1 - \alpha,$$

and choose middle values $\eta_0, \dots, \eta_{M-1}$. Set $\delta_0 = \eta_0 - s$ and $\delta_j = \eta_j - \eta_{j-1}$ for $j > 0$. Define

$$I_\alpha(x) = \int_0^x \frac{v^{\alpha-1}}{1-v}\,dv, \quad K = I_\alpha(\tau), \quad D = I_a(\tau) > 0.$$

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For $t+r<1$ put

$$J_\alpha(t,r)=\int_0^{1-t-r}\frac{v^{\alpha-1}}{1-v}\log\frac{(1-t-v)(1-r-v)}{tr}\,dv,$$

and put $J_\alpha(t,r)=0$ otherwise. Positive real powers use real logarithms. This also equals the integral of $(1-u-v)^{\alpha-1}/(uv)$ over $u\ge t$, $v\ge r$, $u+v\le 1$. Finally set

$$Y=1+sK-\sum_j \delta_j(I_\alpha(1-t_j)-K)+\frac{1}{2}\sum_{i,j}\delta_i\delta_j J_\alpha(t_i,t_j). \tag{1}$$

Proposition 2. *If $|s|<C$, $|\eta_j|<C$ for all j, and $|Y|<CD$, then $R_n\le C$ for every sufficiently large integer n. Consequently $C_{42}\le C$.*

Proof. Put $A=\lfloor n/3\rfloor$. Split the indices into $E=\{1,\dots,A\}$, $M_n=\{A+1,\dots,n-A-1\}$ and $F=\{n-A,\dots,n\}$. On E set $S_m=s$. On M_n put

$$w_m=\sum_j \delta_j\mathbf{1}_{m>\lfloor nt_j\rfloor},\qquad S_m=s+w_m.$$

Every such value is one of the η_j. Write $\beta_l=(\alpha)_l/l!$ and $P_m=n\beta_{n-m}/m$ on F. These numbers are nonzero because $a>0$. Set

$$Y_n=n\beta_n-n\sum_{m\in M_n}\frac{w_m\beta_{n-m}}{m}+\frac{n}{2}\sum_{\substack{u,v\in M_n\\ u+v\le n}}\frac{w_uw_v\beta_{n-u-v}}{uv}+s\sum_{m\in F}P_m,$$

$$S_m=\frac{Y_n}{\sum_{r\in F}|P_r|}\frac{\overline{P_m}}{|P_m|}\qquad(m\in F).$$

Consider coefficients b_l defined by

$$\mathcal{B}(z)=\frac{1}{1-z}\exp\left(-\sum_{m=1}^{n}\frac{S_mz^m}{m}\right)=\sum_{l\ge 0}b_lz^l.$$

Relative to $(1-z)^{-\alpha}$ every correction index exceeds A. Since $3(A+1)>n$, at most two corrections contribute to b_n. A final correction plus any other correction has index at least $n+1$. Expanding the exponential therefore gives exactly $nb_n=Y_n-\sum_{m\in F}S_mP_m=0$.

Let $y_2,\dots,y_n$ be the roots of $Z^{n-1}+b_1Z^{n-2}+\cdots+b_{n-1}$, with multiplicity, and let $y_1=1$. The logarithmic derivative of $\mathcal{B}$ and Newton's identities give the same triangular recurrence for S_k and $Q_k=\sum_j y_j^k$ through $k=n$; at the last step use $b_n=0$. Hence $Q_k=S_k$ for $1\le k\le n$. Dividing all roots by $\Lambda=\max_j|y_j|\ge 1$ cannot increase these power-sum moduli.

It remains to bound the final powers. The gamma-ratio asymptotic is

$$\beta_l=\frac{l^{\alpha-1}}{\Gamma(\alpha)}(1+O(l^{-1})),\qquad\sum_{l\le L}|\beta_l|=O((L+1)^a).$$

Away from $l=0$, ordinary Riemann sums give

$$\Gamma(\alpha)n^{-\alpha}Y_n\longrightarrow Y,\qquad|\Gamma(\alpha)|n^{-a}\sum_{m\in F}|P_m|\longrightarrow D.$$

For completeness, discard first $l<\varepsilon n$ in each sum. In a single sum $m\asymp n$. In a double sum group by $l=n-u-v$: there are at most n pairs and $uv\gg n^2$. After the relevant normalization the discarded contribution is $O(\varepsilon^a+n^{-a})$ in both cases. This permits $\varepsilon\downarrow 0$, including the singular endpoint. The finite number of step discontinuities and the integer floors do not affect the limit. Thus $|Y_n|/\sum_F|P_m|\to|Y|/D<C$. All final powers have modulus below C eventually, which proves the assertion for all sufficiently large n, not merely a subsequence. □

3. Rational certificate and arithmetic

Use $M = 256$,

$$t_j = \frac{256 + j}{768} \quad (0 \le j \le 256), \qquad C = 0.6888863712,$$
$$\alpha = 0.64427390171288 + 0.58993506009068i.$$

The 256 pairs η_j are the terminating-decimal rational strings in `certificates/c42_256.json`. The complete certificate, including these data, has SHA-256

`705a729155eb508878d17ebea0322627`
`a0ae02451c044a346a99d2e81c7afcab.`

The file is the explicit definition of the profile; unverified floating search diagnostics retained in it are not inputs to the proof. The exact minimum squared profile margin is

$$\min\{C^2 - |s|^2, C^2 - |\eta_j|^2\} = \frac{1491623179869746287}{500000000000000000000000000} > 0.$$

Outward enclosures, shortened here, give

$$D \in [0.89218420910724133656675082 6518, 0.89218420910724133656675082 6519],$$
$$|Y|/D \in [0.68888637098346564582276780 3129, 0.68888637098346564582276780 3130],$$
$$C^2D^2 - |Y|^2 \in [2.37472575330897264258, 2.37472575330897264259]\, 10^{-10}.$$

Proposition 2 proves Theorem 1.

Two independently authored implementations accept these inputs. The first is Röhrig's unchanged public `mpmath.iv` checker at 100 decimal places, using explicit geometric series tails for I and J. The second, written by an independent agent from the mathematical notes without reading either search code or the first checker, uses 768-bit Arb balls. It computes I through

$$I_\alpha(x) = \frac{x^\alpha}{\alpha} {}_2F_1(1, \alpha; \alpha + 1; x)$$

and J through a separately bounded logarithmic series with 400 terms. No optimization status, sampled integral or floating-point maximum is accepted as a certificate. Corrupted profiles, malformed thresholds and the too-strong radius 0.6888863709 are rejected.

4. Reproduction and limits

The local package includes pinned public source code, both checkers, exact data and transcripts. From its root, with Python, mpmath and python-flint available, run

```
python verify_primary.py certificates/c42_256.json
python independent/check_bounds.py \
  certificates/c42_256.json --bits 768 --terms 400
```

The proof gives existence of an eventual threshold, not its numerical value. The optimizer's final status was a line-search warning with tiny floating constraint residuals; only the subsequent strict interval checks justify the result. Neither the profile nor its resolution is claimed optimal. The independent review is a separate AI-agent review, not independent human expert review or proof-assistant formalization. Final source checks and the exact search commands are recorded in the local `PRIOR_ART.md` and `REPORT.md`.

Acknowledgments and assistance

The method is due to Griego and its step-profile refinements to Röhrig and Plata. Analysis and proof drafting by AI agents under my direction; I reviewed the results at summary level. Search, code, verification and exposition were developed with OpenAI Codex assistance; the second checker was written in a separate agent context. No submission, upload or contact occurred.

References

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